

Applied Statistics Problem Set

Sagar Clemensen – fck916

Discussed problems with: Aske Roelshøj and Rasmus Ahrensberg

03/01 - 2026

1 Distributions and Probabilities

1.1 Distribution of push-ups

The distribution of days between push-ups follows the geometric distribution used for modeling the number of failures until the first success. The probability mass function is

$$P(X = k) = (1 - p)^k p, \text{ for } k = 0, 1, 2, 3, \dots, \quad (1)$$

where X is the amount of failures or days between push-ups and $p = 1/6$. I also simulated the distribution across 1000000 days and plotted this alongside the geometric pmf which can be seen in fig. 1.

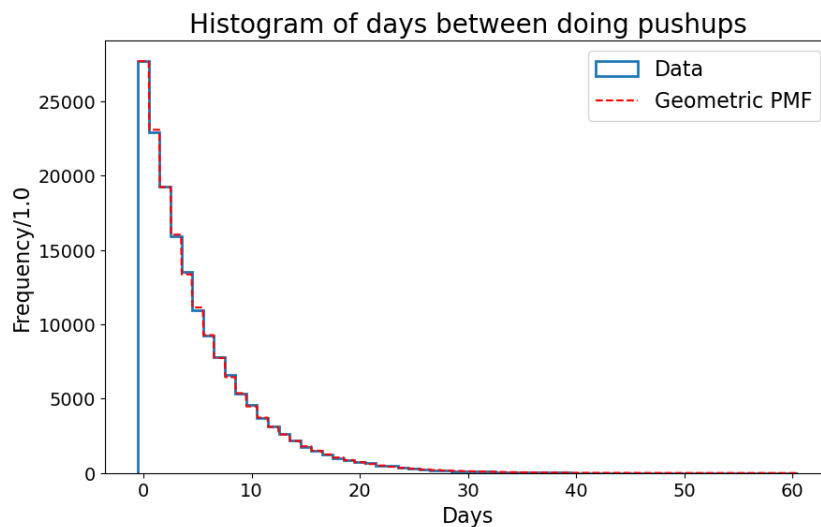


Figure 1: Analytical and simulated distributions. The simulation is of 1000000 initial die rolls/days.

Splitting the rolls of the die into 10-day intervals the distribution seen in fig. 2 is obtained. The mean, median and standard deviation can be seen in table 1 with the she standard error on the mean (SEM) being used. Since the standard deviation is an estimate of the true distribution it is also given with its error.

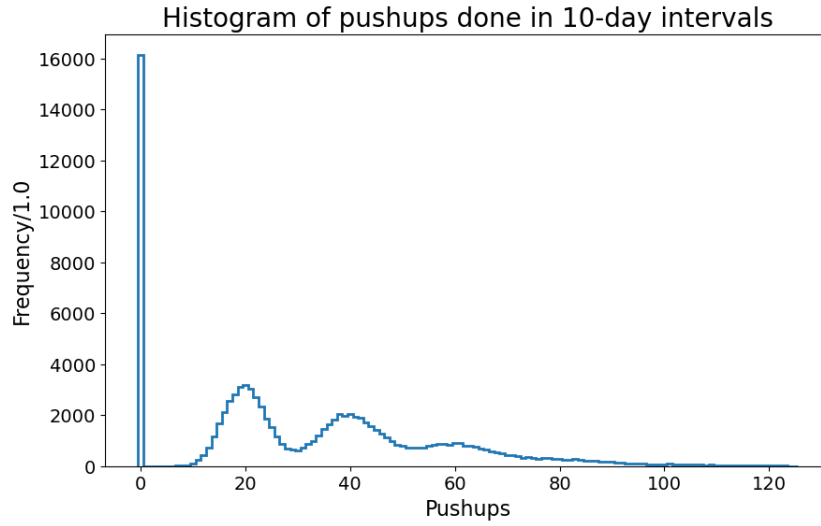


Figure 2: Simulated distribution of push-ups across 100000 10-day intervals.

Mean	Median	Standard deviation
33.24 ± 0.08	31	24.11 ± 0.05^a

Table 1: Results from distribution of push-ups in problem 1.1.

^a The error on the standard deviation is not taken into account when calculating the SEM.

1.2 Djoser pyramid

The base of the pyramid is assumed to be a square resulting in a rotational symmetry of order 4. This gives it a unique rotational range of 90° . The pyramid is North-South aligned to $\pm 3^\circ$ which is a rotational range of 6° . Hence the probability that the Djoser pyramid is North-South aligned by coincidence is

$$P = \frac{6}{90} = \frac{1}{15}. \quad (2)$$

2 Error Propagation

2.1 Water origin

If f is the fraction of water that comes from the original proto-solar system and $1 - f$ is the fraction from comets, then the water on Earth must satisfy the following equation for its Deuterium to Hydrogen ratio

$$r_{\oplus} = fr_{\odot} + (1 - f)r_C. \quad (3)$$

This can be rearranged to obtain an equation for the fraction f

$$f = \frac{r_{\oplus} - r_C}{r_{\odot} - r_C}. \quad (4)$$

Using error propagation it is possible to calculate the central value of f and its uncertainty which was done using Sympy. Since the fraction of water that comes from comets is $1 - f$ the uncertainty is the same as for f . The values of f and $1 - f$ can be seen in table 2.

Fraction of water from the original proto-solar system f	Fraction of water from comets $1 - f$
0.56 ± 0.03	0.44 ± 0.03

Table 2: Results from error propagation in problem 2.1.

2.2 Detector rate

The error on the amount of counts from the detector σ_N is $\sqrt{N}$. Using error propagation it is then possible to calculate the rate r and its uncertainty σ_r , which is again done using Sympy. The resulting rate is

$$r = (20.0 \pm 0.9)s^{-1}, \quad (5)$$

with a relative uncertainty of 4.4%. Assuming the rate for the first interval to be constant, running the detector for a time interval $\Delta t = 193.2s$ yields $N = 3869$ counts. This results in the rate

$$r = (20.0 \pm 0.5)s^{-1}, \quad (6)$$

with a relative uncertainty of 2.5%. Running it for an additional 0.1s would then give a relative uncertainty below 2.5%. So you would have to measure for a time interval of $\Delta t = 193.3s$ to obtain a relative uncertainty on r below 2.5%.

2.3 Pylon positions

I determined the pylon positions from the measurements without reported uncertainties using two independent methods. The data set consists of 120 unlabeled measurements. After sorting the values in ascending order, I assumed that the first 30 measurements correspond to pylon 1, the next 30 to pylon 2, and so forth. I then used Chauvenet's criterion to reject some of the measurements. These are shown in Appendix A fig. 24. For each group of accepted measurements I calculated the mean, standard deviation, and the standard error on the mean (SEM).

As an alternative approach, I performed an unbinned likelihood (ULLH) fit on the sorted data without outliers using a sum of four Gaussian pdf's. The fit parameters were optimized using Minuit, and the mean of each Gaussian peak was interpreted as the corresponding pylon position. The resulting fit is shown in fig. 3 and the results are identical to those obtained from averaging.

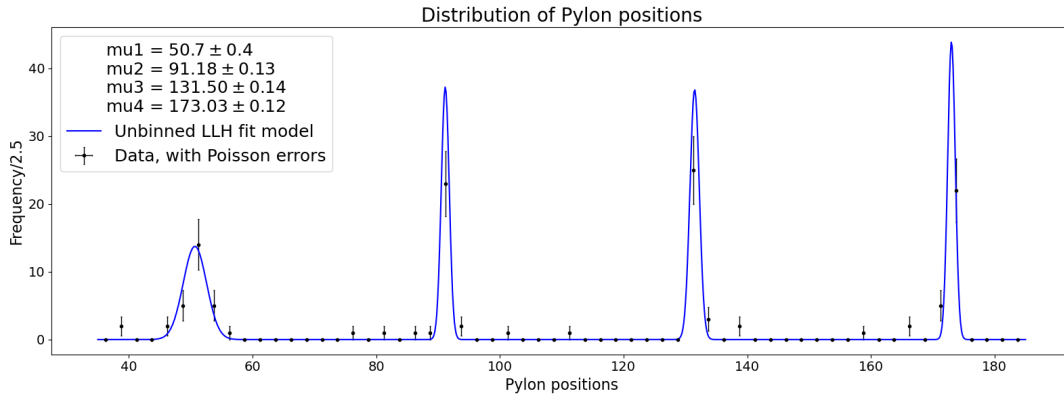


Figure 3: The ULLH fit is performed with minuit.

	Pylon 1	Pylon 2	Pylon 3	Pylon 4
Average	50.7 ± 0.4	91.18 ± 0.13	131.50 ± 0.14	173.03 ± 0.12
ULLH fit	50.7 ± 0.4	91.18 ± 0.13	131.50 ± 0.14	173.03 ± 0.12

Table 3: Positions of pylon 1-4 calculated with average of values and ULLH fit. The results are indeed identical.

For the measurements with uncertainties 80 of them had an invalid uncertainty of -1 and the corresponding measurements were thus discarded. A overview of all valid measurements is provided in Appendix A fig. 26. Using the previously obtained pylon positions as reference values, I calculated a z-value for each valid measurement. Measurements with $|z| > 2$ were excluded to reduce the influence of outliers. The remaining measurements used for each pylon are shown in Appendix A fig. 25. Using these I calculated a weighted mean for each pylon position. In addition, a χ^2 value and p-value were computed to evaluate the consistency of the measurements with the weighted mean. The results are shown in table 3.

Since the lowest p-value is 0.35, none of the χ^2 tests indicate a statistically significant difference. Therefore, the weighted means are consistent with the individual measurements and provide a reliable estimate of the pylon positions based on measurements with uncertainties.

Values	Pylon 1	Pylon 2	Pylon 3	Pylon 4
Position	50.6 ± 0.6	90.7 ± 0.3	131.3 ± 0.3	173.2 ± 0.3
χ^2	8.7	5.6	7.8	6.4
N_{dof}	8	6	7	8
p	0.37	0.48	0.35	0.60

Table 4: Positions of pylon 1-4 calculated using the weighted mean of measurements with uncertainty. p-values show whether values and the weighted mean agree.

The comparison of the measurements groups was performed by calculating z-values and corresponding (two-sided) p-values. As shown in table 5, all p-values are well above a significance threshold of 0.05, indicating that the two measurement groups are statistically compatible. I therefore combined the two groups using a weighted mean. The final combined pylon positions are also listed in table 5.

	Pylon 1	Pylon 2	Pylon 3	Pylon 4
z	0.17	1.46	0.51	0.58
p	0.87	0.14	0.61	0.56
Final position	50.7 ± 0.3	91.1 ± 0.1	131.5 ± 0.1	173.1 ± 0.1

Table 5: (Two-sided) p-values from z-tests comparing pylon positions obtained from measurements without uncertainties and with uncertainties. The final combined pylon positions are also listed.

The distance between each pylon was calculated as $Pos_{i+1} - Pos_i$ for $i = 1, 2, 3$. The uncertainty on each distance was obtained by propagating the uncertainties on the individual pylon positions according to

$$\sigma_{dist} = \sqrt{\sigma_i^2 + \sigma_{i+1}^2}, \text{ for } i = 1, 2, 3, \quad (7)$$

where σ_i denotes the uncertainty of the position of pylon i . The resulting distances are shown in table 6.

To test whether the pylons are equidistant, I performed two statistical tests. First, I calculated the weighted mean of the three distances and computed a χ^2 and p-value. Secondly, I calculated pairwise z-values and (two-sided) p-values between the distances. The p-values obtained from both the χ^2 test and the pairwise comparisons are summarized in table 7. Since the p-values from the weighted mean and the pairwise between distance 1/2 and 3 are 0.00 there is statistically significant evidence against the hypothesis that the pylons are equally spaced.

Pylon 1 and 2	Pylon 2 and 3	Pylon 3 and 4
40.4 ± 0.3	40.4 ± 0.2	41.6 ± 0.2

Table 6: Distances 1-3 between pylon positions.

Weighted mean	Distance 1 and 2	Distance 1 and 3	Distance 2 and 3
0.00	0.85	0.00	0.00

Table 7: p-values for the weighted mean of distances between the pylons and for each distance pairwise with distance 1 being between pylon 1 and 2 and so forth.

3 Simulation/Monte Carlo

3.1 Overlapping Balls

For both 2D and 4D I used the accept/reject method to simulate balls A and B with 100000 points inside each and then denoted how many where overlapping and calculated the subsequent fraction. For the 2D example I also plotted it which can be seen in fig. 4. In table 8 the results from the simulation can be seen.

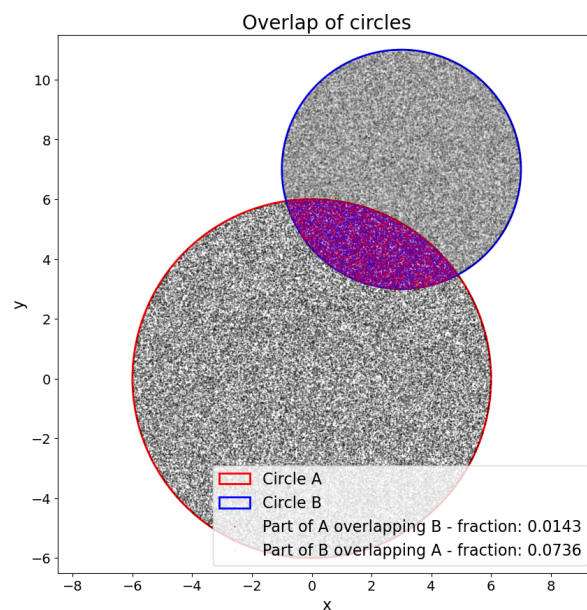


Figure 4: 2D circles with overlapping points marked

Dimension	Fraction of A overlapping with B	Fraction of B overlapping with A
2D	0.0910	0.2045
4D	0.0143	0.0736

Table 8: Results from accept/reject method in problem 3.1.

3.2 Radial material distributions

The distributions of r , θ , and ϕ can be seen in fig. 5, 6, and 7.

In order to produce random velocities v according to $f(v) = (v/v_0)^2 \exp(-v/v_0)$ I used the accept reject method, since the transformation method required inverting the function $1 - (v_0 v^2 + 2v_0 v + 2v_0^3) \exp(-v/v_0) / 2v_0^3$. Even though $f(v)$ doesn't have a limit along the x -axis it does go to zero at some point since it is an exponentially decreasing function. In fig. 8 the simulated velocities are shown together with the normalized velocity function $f_N(v) = (1/2v_0)(v/v_0)^2 \exp(-v/v_0)$. For the normalization it is assumed that the velocities are bounded as $v \in [0, \infty[$. I decided to simulate 13011 values of v since that was the amount of points that obeyed $z > 0$ and $r < 0$ and thereby the amount of simulated θ values.

The resulting radial material distribution $m(r)$ is seen in fig. 9. A logarithmic y-axis is used.

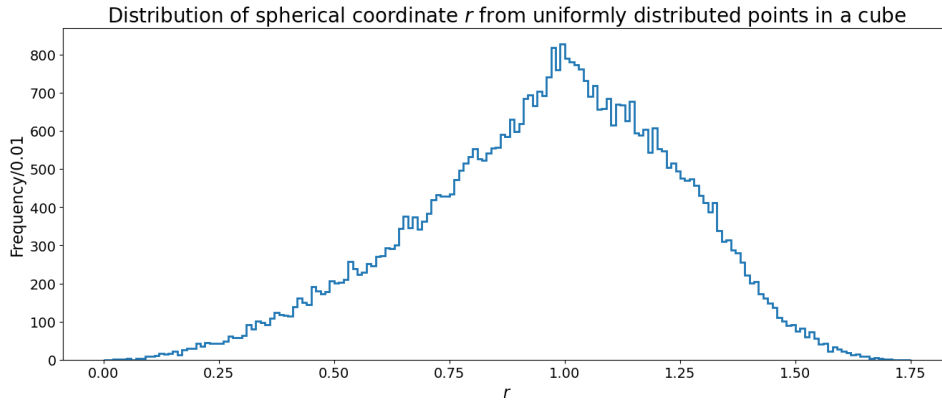


Figure 5: Distribution of the spherical coordinate r from 50000 randomly generated x , y , and z values in the range $[-1, 1]$.

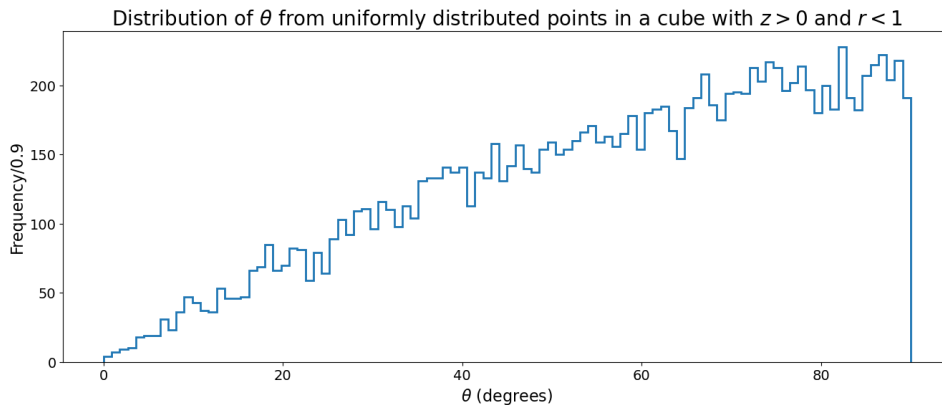


Figure 6: Distribution of θ . There are 13321 entries.

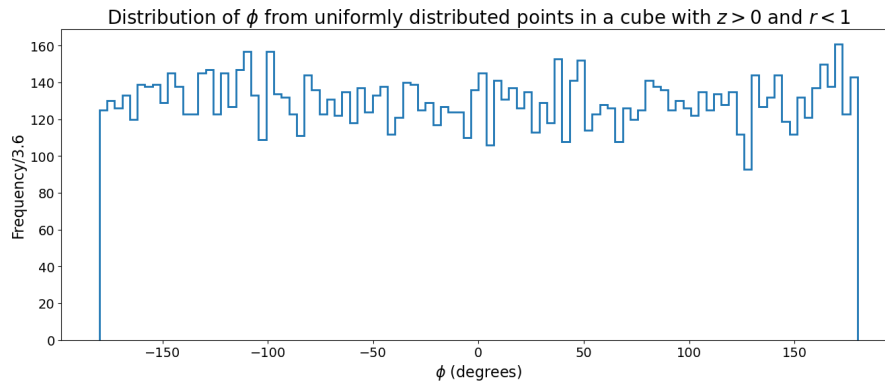


Figure 7: Distribution of ϕ . There are 13321 entries.

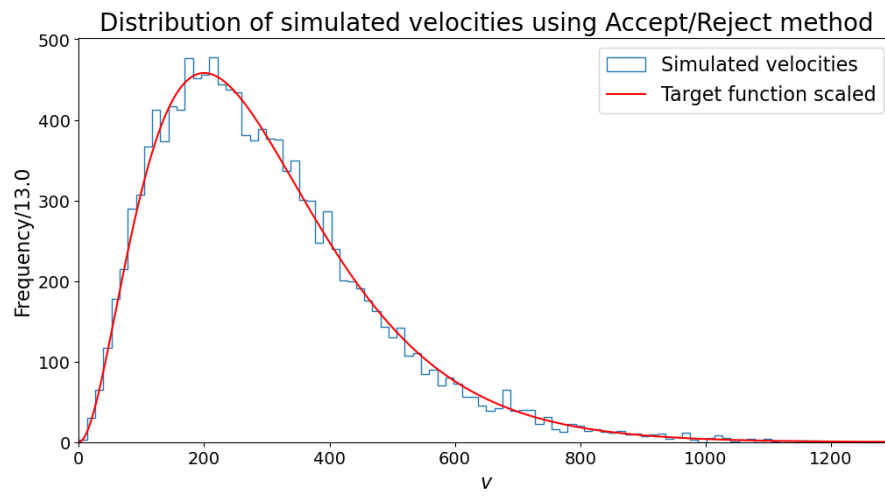


Figure 8: Distribution of simulated velocities. There are 13321 entries.

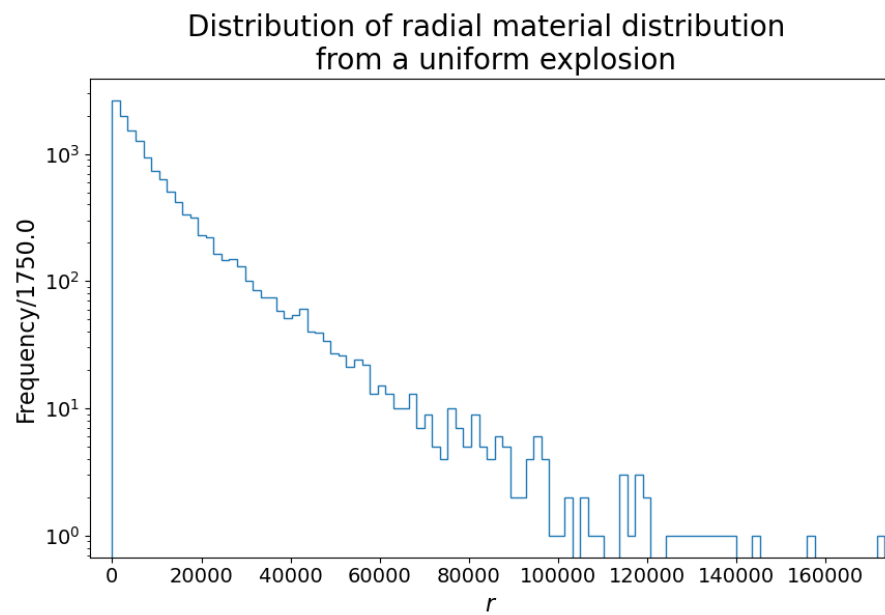


Figure 9: Distribution of $m(r)$. A logarithmic y-axis is used since the entries are spread out across a large range of values.

4 Statistical tests

In fig. 11 the die frequencies from 200 rolls of a box with 25 dies are plotted. The one dataset has only real die while the other has one fake die. The data in this plot is blinded as to not disclose which one has the fake die. A Pearson's χ^2 and resulting p -value is calculated to test whether there is a fake die. Dataset B clearly fails the test as it has a p -value of 0.000, so it should have the fake die. Looking at the unblinded plot in fig. 11 it is indeed dataset B that has the fake die.

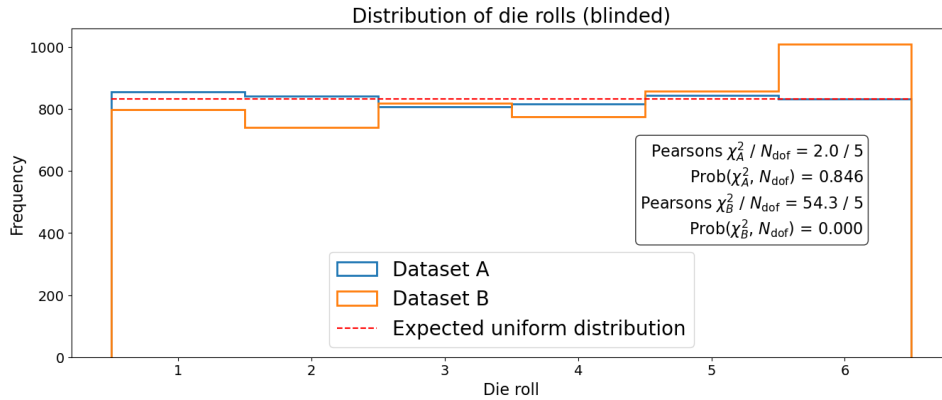


Figure 10: Blinded

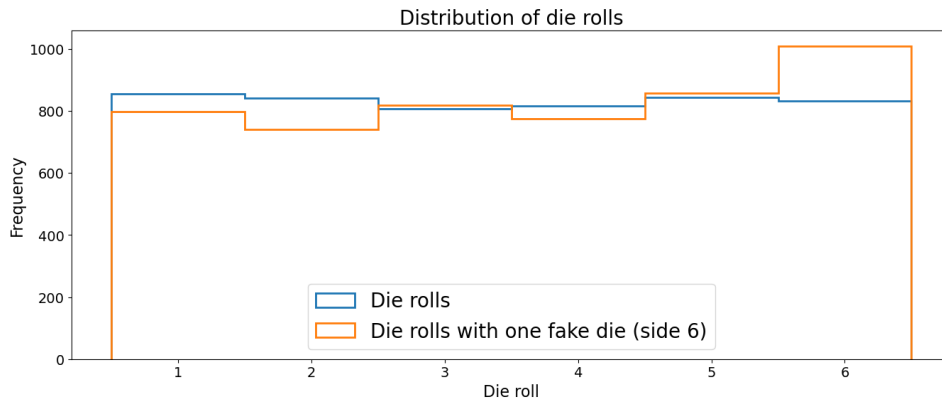


Figure 11: Unblinded

In order to determine how many rolls are necessary before you can tell if a box has a fake die I devised a test where you trace the (χ^2) p -value after every roll of the box. I then had two conditions; either the difference in p -values was greater than 0.1 or the p -value of the box with a fake die was lower than 0.05. If either of these were met for 10 consecutive rolls I argue that at the current amount of rolls the box with a fake die can be determined. This was done for 1000 different seeds and then plotted in a histogram fig. 12. On a 0.05 significance level 70 die rolls are needed to determine whether a box has a fake die. On a 0.01 significance level 100 die rolls are needed. It is also possible to do this test qualitatively by examining the p -value traces. Such a plot can be seen in

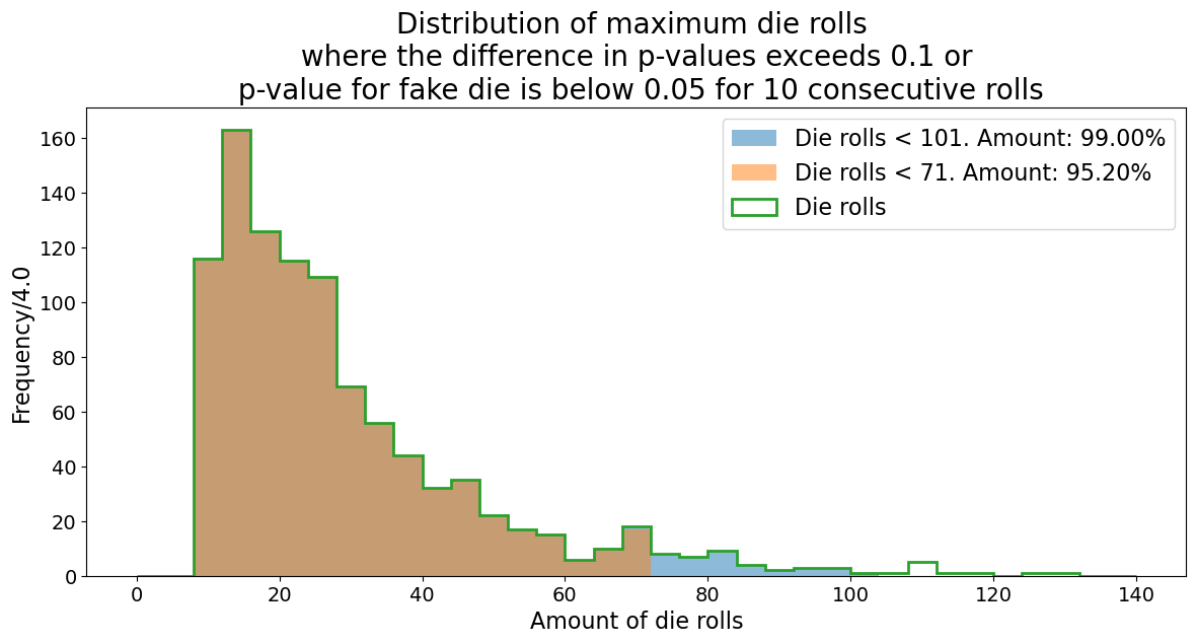


Figure 12: Distributions of amount of rolls needed before either condition is met 10 consecutive times.

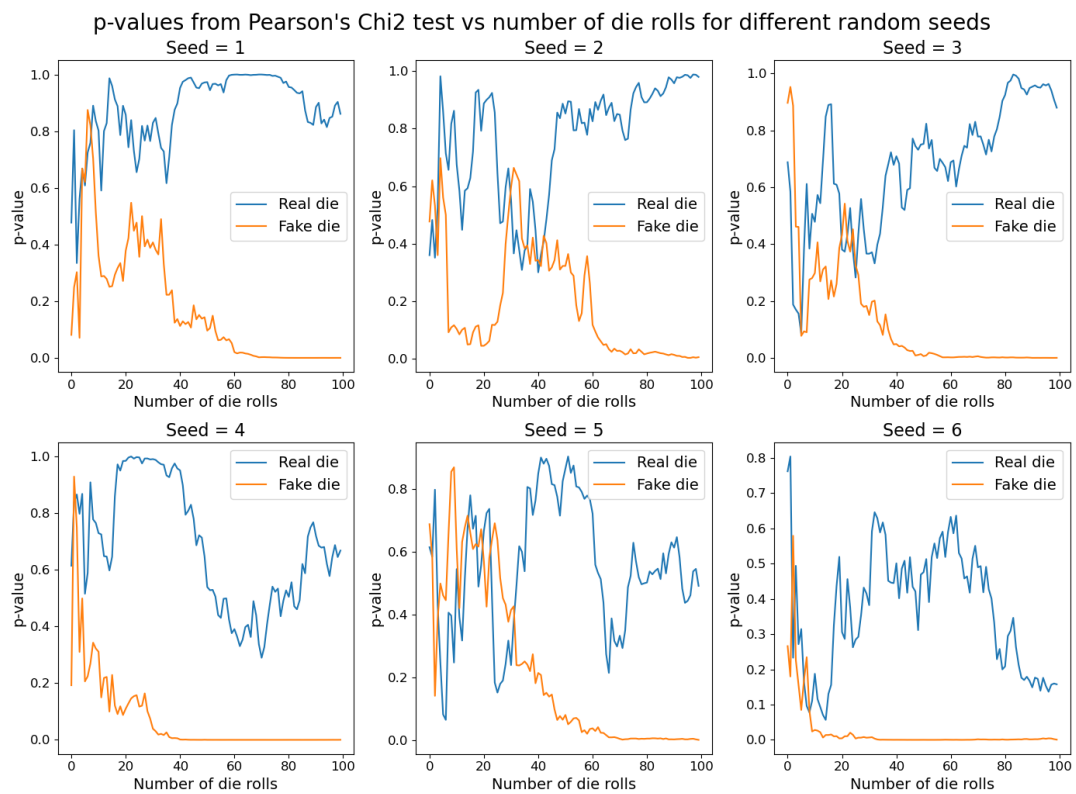


Figure 13: The trace of p-values for a box containing a fake die and one with only real dies for 6 different seeds.

5 Fitting Data

5.1 Molecular interspacing measurements

The data from the scattering measurement is plotted in fig. 14. In fig. 15 the background in the range $[8, 10]$ is plotted alongside a sample from a true uniform distribution. In order to determine whether the background is uniform a Kolmogorov-Smirnov test is performed. As it yields a p-value of 0.18 it is not sufficient to reject the hypothesis that the background in the range $[8, 10]$ is uniform.

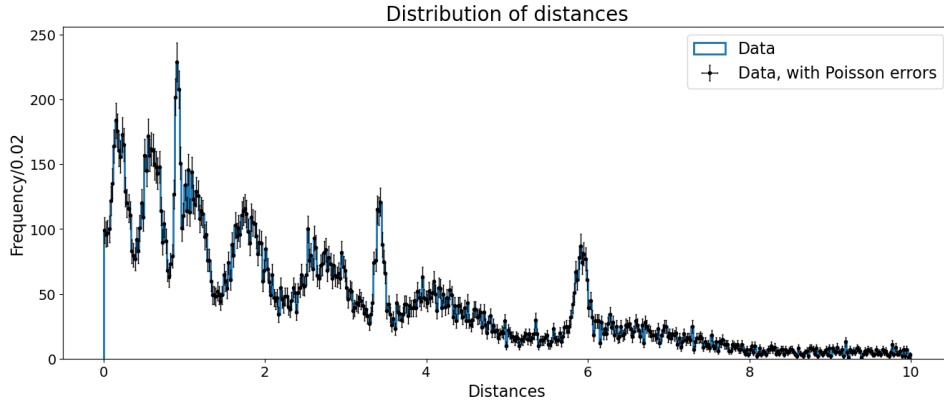


Figure 14: Distribution of molecular interspacing measurements d (in nm) from a scattering experiment.

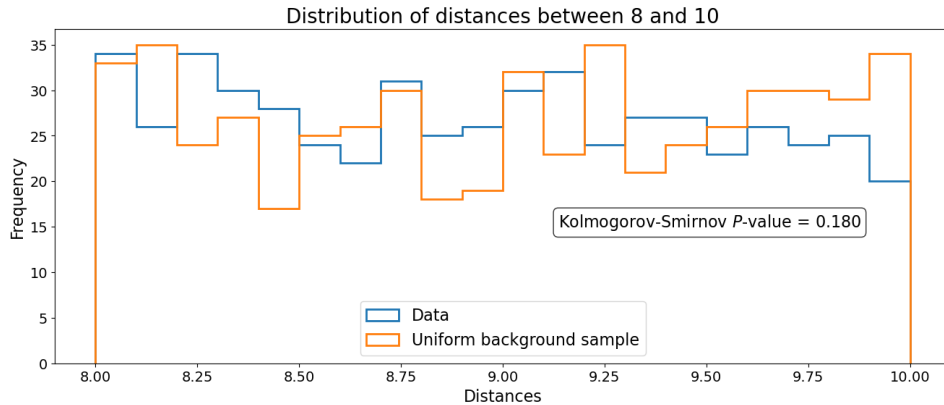


Figure 15: Plot of background from scattering experiment in the range $[8,10]$ together with a sample from a true uniform distribution.

Each peak is fitted locally with Minuit. Peak 1 is fitted together with a linear background, peak 2 with a parabolic, and peak 3 with a Gaussian. A χ^2 -fit is used since all bins have reasonable statistics (90% of bins have more than 5 counts). The fits are shown in fig. 16, 17, and 18.

I calculated the weighted mean of the three intensities and computed a χ^2 value and corresponding p-value and calculated pairwise z-values and p-values between the intensities. The p-values obtained from both the χ^2 test and the pairwise comparisons are summarized in table 9. Since the lowest p-value from the weighted mean and pairwise is 0.07 there is not statistically significant evidence against the hypothesis that the peaks have the same intensity.

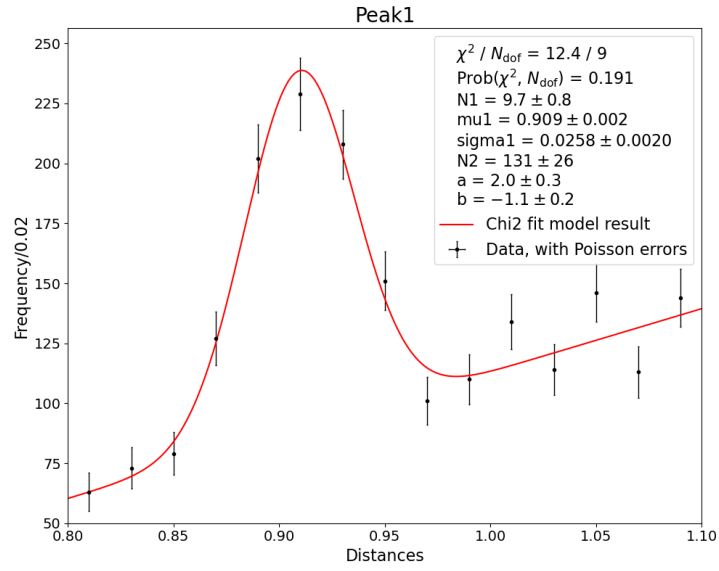


Figure 16: χ^2 fit of a Gaussian peak with linear background

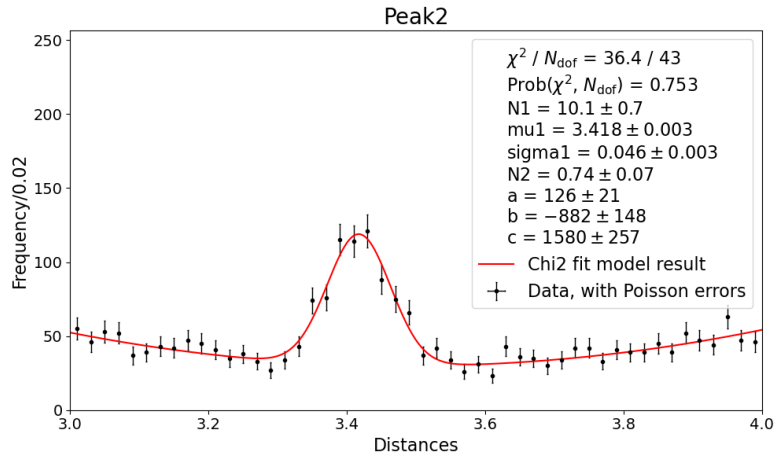


Figure 17: χ^2 fit of a Gaussian peak with parabolic background

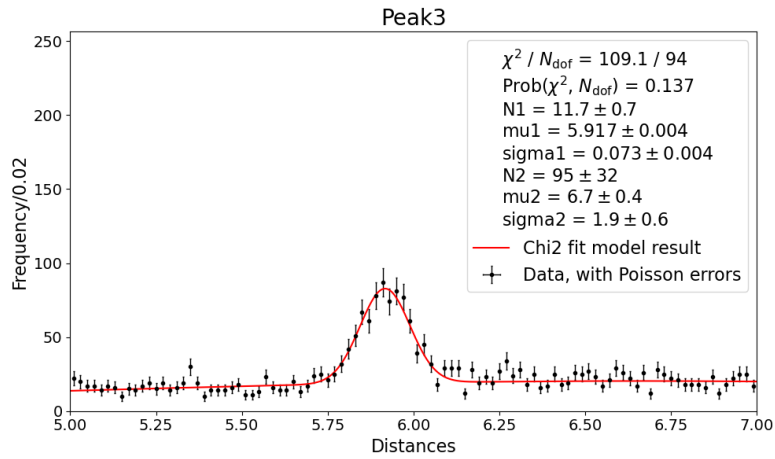


Figure 18: χ^2 fit of a Gaussian peak with Gaussian background

Weighted mean	Peak 1 and 2	Peak 1 and 3	Peak 2 and 3
0.14	0.65	0.12	0.07

Table 9: p-values for the weighted mean of intensities of peaks in distance measurements and for each intensity pairwise.

After having fit the three peaks I subtracted them from the data to have just background. I then fitted the background with two different functions. The first consisted of 7 Gaussian, while the second was a modified sine function:

$$f(x) = N_{exp}e^{-x/\tau} + e^{-x\beta}A\left(\frac{\omega}{1+\alpha x}x + \phi\right). \quad (8)$$

The Gaussian fit is shown in fig. 19 and the sine in fig. 20. Finally the two background fits are added together with a triple Gauss for the three peaks and then a full fit to all the data is done. The full fit with Gaussian background is shown in fig. 21 and the sine in fig. 22.

The χ^2 , N_{dof} , and p -values for the different fits are listed in table 10. Fitting the background with 7 Gaussians yields a lower χ^2 both with and without the peaks, but due to its large number of free parameters it carries an increased risk of overfitting compared to the simpler eq. (8). The sine-model also gives a higher p-value both with and without the peaks. The sine fit is also way faster to compute.

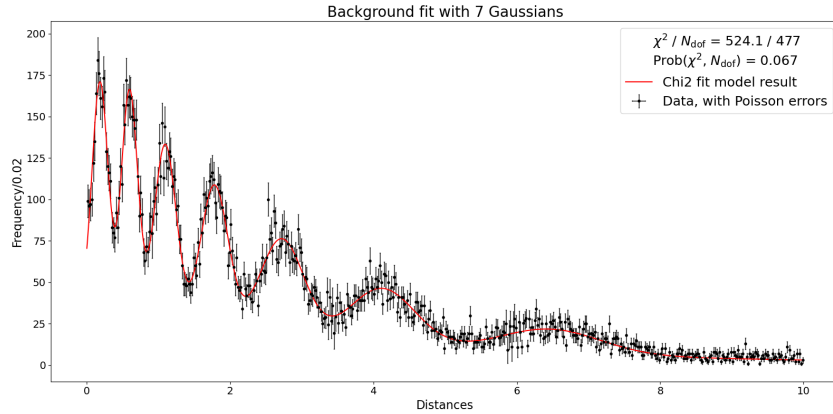


Figure 19: χ^2 fit with 7 Gaussian to background.

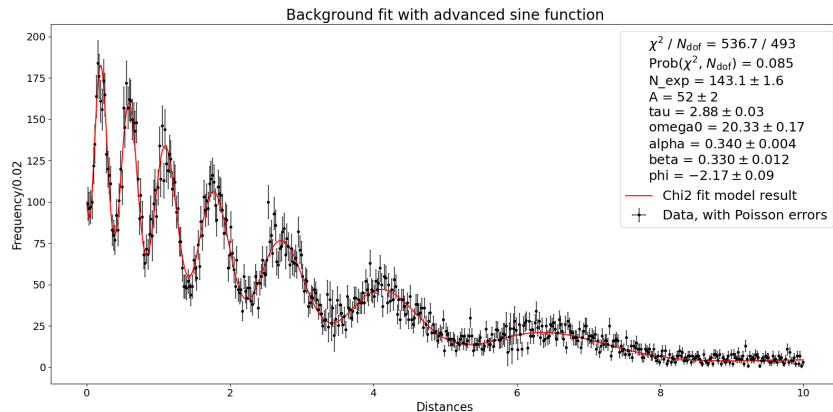


Figure 20: χ^2 fit with eq. (8) to background.

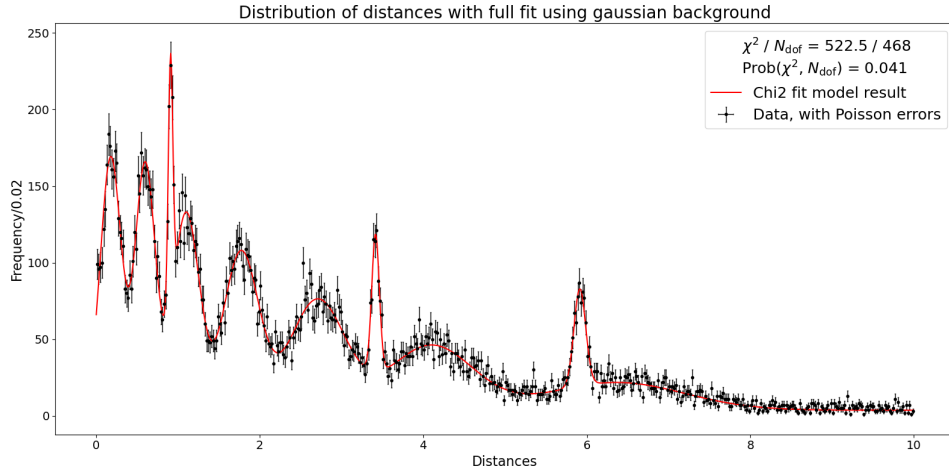


Figure 21: χ^2 fit of the three Gaussian peaks with Gaussian background.

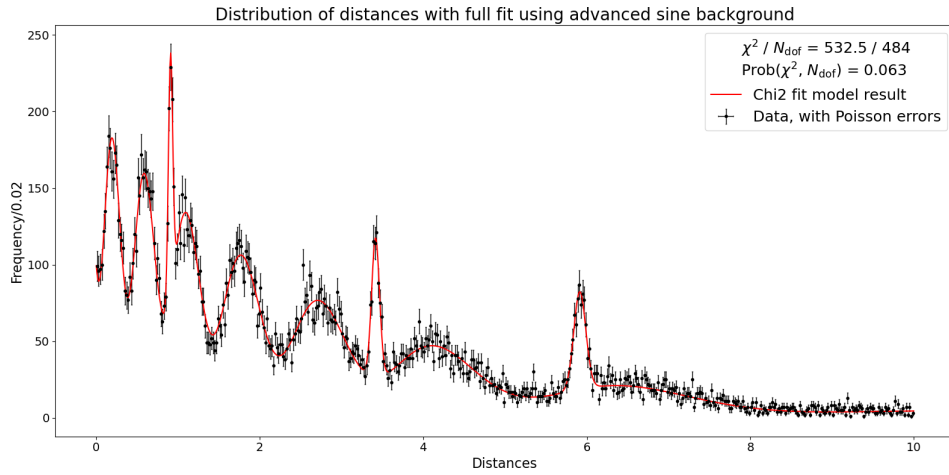


Figure 22: χ^2 fit of the three Gaussian peaks sine background eq. (8).

Fit	χ^2	N_{dof}	p
Bkg - Gauss	524.1	477	0.067
Bkg - sine	536.7	493	0.085
Full data - Gauss bkg	522.5	468	0.041
Full data - sine bkg	532.5	484	0.063

Table 10: χ^2 , N_{dof} , and p -value for the different fits to just background and peaks + background.

5.2 Pyramid alignments

Looking at table 11 the alignments for Meidum and Menk agree at a significance level of 0.05. Red, Sahure, and Nefer are given with only an east alignment value and so it doesn't make sense to argue whether they agree. To account for unreported systematic uncertainties, additional systematic errors were introduced to the pyramids Bent and Khufu in increments of 0.1 to ensure agreement at a significance level of

0.05. After this the values were combined with a weighted mean. The new systematic uncertainties, z- and p-values as well as combined alignments can be seen in table 12. To determine whether the alignment shifts linearly as a function of time a χ^2 linear fit is performed using Minuit and shown in fig. 23. This gives $\chi^2 = 28.4$ and $p(28.4, 6) = 0.000$ which is sufficiently low to reject the hypothesis that the alignments of the pyramids shifts linearly with time.

The slope of the fit is (0.295 ± 0.004) *arc min./year*. Comparing this to the astronomically predicted value of 0.274 *arc min./year* yields $z = 4.95$ and $p = 0.00$ which is sufficiently low to reject the hypothesis that the two values match.

Assuming that the pyramid alignments vary linearly with time, using the fitted slope from fig. 23 to extrapolate to Khufu's deviation from true north yields an alignment date of 2477.5 ± 0.7 BC for Khufu. The quoted uncertainty is obtained by propagating the errors in both the slope and Khufu's combined alignment.

Pyramid	Meidum	Bent	Red	Khufu	Khafre	Menk	Sahure	Nefer.
z	1.77	19.45	—	2.12	0.00	0.83	—	—
p	0.08	0.00	—	0.03	1	0.41	—	—

Table 11: z- and p-values for agreement between East and West alignment for the pyramids. Three of the pyramids (Red, Sahure and Nefer.) are given with only an East alignment and are therefore not included here.

Pyramid	Meidum	Bent	Khufu	Khafre	Menk
East	-20.6 ± 1.0	$-17.3 \pm 0.2 \pm \mathbf{1.8}$	$-3.4 \pm 0.2 \pm \mathbf{0.1}$	6 ± 0.2	12.4 ± 1.0
West	-18.1 ± 1.0	$-11.8 \pm 0.2 \pm \mathbf{1.8}$	$-2.8 \pm 0.2 \pm \mathbf{0.1}$	6 ± 0.2	14.1 ± 1.8
z	1.77	1.94	1.41	0.00	0.83
p	0.08	0.05	0.16	1	0.41
Comb.	-19.4 ± 0.7	-14.6 ± 1.4	-3.1 ± 0.2	6 ± 0.2	12.8 ± 0.9

Table 12: Alignment with systematic uncertainties (marked with bold), z- and (two-sided) p-values, and combined alignment values given in arc minutes. Once again pyramids Red, Sahure and Nefer. are not included.

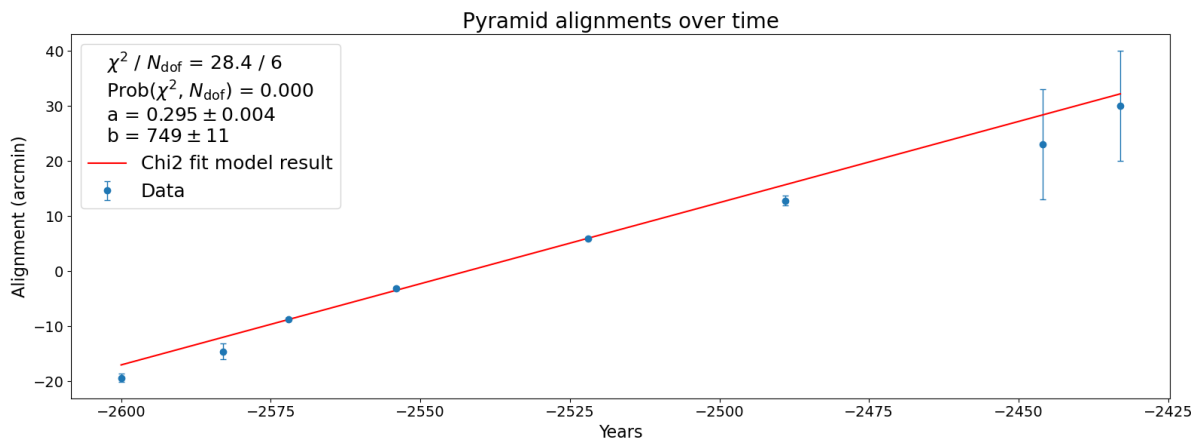


Figure 23: Linear fit of pyramid alignments over time minimizing χ^2 with Minuit. The p-value is sufficiently low to reject the hypothesis that the pyramid alignments shift linearly with time.

A Pylon Positions

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Pylon 1: Rejected 4 out of 30 measurements: [75.341 38.946 39.348 57.19 ]
with z-values [4.20810811 3.03981505 2.93047714 2.81624715]

Pylon 2: Rejected 6 out of 30 measurements: [ 80.364 110.35 102.12 86.333 93.459 89.421]
with z-values [2.45213554 4.04784557 4.48099485 3.80220852 2.70429174 2.34921886]

Pylon 3: Rejected 2 out of 30 measurements: [137.712 139.577]
with z-values [2.95466817 3.91521344]

Pylon 4: Rejected 3 out of 30 measurements: [159.91 165.978 166.262]
with z-values [4.20372722 3.55718408 3.40350232]
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Figure 24: The measurements removed by Chauvenet's criterion

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Pylon 1: Using 9 out of 120 measurements after validity check and pull criterion
Valid measurements: [47.813 53.014 50.852 50.391 48.16 53.047 50.831 49.171 51.958]
and uncertainties: [1.83 1.831 1.821 1.874 1.838 1.856 1.841 1.863 1.881]

Pylon 2: Using 7 out of 120 measurements after validity check and pull criterion
Valid measurements: [91.403 90.068 91.008 90.411 90.275 90.463 92.582]
and uncertainties: [1.162 0.676 0.74 0.675 0.804 0.81 0.993]

Pylon 3: Using 8 out of 120 measurements after validity check and pull criterion
Valid measurements: [132.792 130.587 131.47 131.74 131.612 131.737 130.139 130.86 ]
and uncertainties: [0.818 0.846 0.673 0.929 0.771 0.671 0.727 0.741]

Pylon 4: Using 9 out of 120 measurements after validity check and pull criterion
Valid measurements: [172.356 173.611 172.996 173.446 172.586 173.253 173.993 172.37 174.054]
and uncertainties: [0.681 0.979 1.052 0.746 0.672 1.118 0.722 0.889 0.67 ]
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Figure 25: Measurements with uncertainties used for determining the pylon positions. Only valid measurements with $z < 2$ are used.

1	The entry: Pylon position = 47.81 ± 1.83 was considered valid
2	The entry: Pylon position = 53.01 ± 1.83 was considered valid
3	The entry: Pylon position = 50.85 ± 1.82 was considered valid
4	The entry: Pylon position = 50.39 ± 1.87 was considered valid
5	The entry: Pylon position = 48.16 ± 1.84 was considered valid
6	The entry: Pylon position = 53.05 ± 1.86 was considered valid
7	The entry: Pylon position = 50.83 ± 1.84 was considered valid
8	The entry: Pylon position = 75.34 ± 1.91 was considered valid
9	The entry: Pylon position = 49.17 ± 1.86 was considered valid
10	The entry: Pylon position = 51.96 ± 1.88 was considered valid
11	The entry: Pylon position = 91.40 ± 1.16 was considered valid
12	The entry: Pylon position = 90.07 ± 0.68 was considered valid
13	The entry: Pylon position = 110.35 ± 0.68 was considered valid
14	The entry: Pylon position = 91.01 ± 0.74 was considered valid
15	The entry: Pylon position = 90.41 ± 0.68 was considered valid
16	The entry: Pylon position = 90.28 ± 0.80 was considered valid
17	The entry: Pylon position = 90.46 ± 0.81 was considered valid
18	The entry: Pylon position = 92.58 ± 0.99 was considered valid
19	The entry: Pylon position = 86.33 ± 0.75 was considered valid
20	The entry: Pylon position = 102.12 ± 1.10 was considered valid
21	The entry: Pylon position = 132.79 ± 0.82 was considered valid
22	The entry: Pylon position = 130.59 ± 0.85 was considered valid
23	The entry: Pylon position = 131.47 ± 0.67 was considered valid
24	The entry: Pylon position = 131.74 ± 0.93 was considered valid
25	The entry: Pylon position = 133.06 ± 0.67 was considered valid
26	The entry: Pylon position = 139.58 ± 0.75 was considered valid
27	The entry: Pylon position = 131.61 ± 0.77 was considered valid
28	The entry: Pylon position = 131.74 ± 0.67 was considered valid
29	The entry: Pylon position = 130.14 ± 0.73 was considered valid
30	The entry: Pylon position = 130.86 ± 0.74 was considered valid
31	The entry: Pylon position = 172.36 ± 0.68 was considered valid
32	The entry: Pylon position = 173.61 ± 0.98 was considered valid
33	The entry: Pylon position = 173.00 ± 1.05 was considered valid
34	The entry: Pylon position = 159.91 ± 0.68 was considered valid
35	The entry: Pylon position = 173.45 ± 0.75 was considered valid
36	The entry: Pylon position = 172.59 ± 0.67 was considered valid
37	The entry: Pylon position = 173.25 ± 1.12 was considered valid
38	The entry: Pylon position = 173.99 ± 0.72 was considered valid
39	The entry: Pylon position = 172.37 ± 0.89 was considered valid
40	The entry: Pylon position = 174.05 ± 0.67 was considered valid

Figure 26: Measurements with valid uncertainties.